

er consumption. The bus clock runs during state transition between the core clocks, and phase-locked loops are stopped while the logic remains active. The core clock can also restart more quickly under SpeedStep than under previous architectures.

Ultra-low-power MCUs for wearable devices

A lot of power is also being saved by reducing consumption in MCUs. The requirement is to have processors that can increase battery life. This has led to a lot of processors with power consumption in $\mu\text{A}/\text{MHz}$, and it is further reducing. STMicroelectronics' STM32L496 and STM32L4A6, for example, work at $37\mu\text{A}/\text{MHz}$. Kinetis L series MCU from NXP work at about $75\mu\text{A}/\text{MHz}$.

Among the wide range of ultra-low-power MCUs, however, ARM Cortex-M processor seems to be the common link. Certain modifications are added to the underlying Cortex core to further improve consumption. For instance, a bit manipulation engine improves time and code size in the architecture.

Kinetis from NXP employs it to the highest effect. Low-power boot is another power-saving option where battery chemistry limits allowable peak currents. The L series allows the peripherals to operate autonomously in deep-sleep mode with an alternate-power source without involving the core or main system.

The ultra-low-power MCU

Apollo 2 integrates up to 1MB flash with 256kB RAM to accommodate radio and sensor overhead. The ARM Cortex-M based processor has a power consumption of under $10\mu\text{A}/\text{MHz}$. It is an improvement over Apollo 1, with current usage of about $35\mu\text{A}/\text{MHz}$ with sleep modes as low as 143nA .

MAX32630/MAX32631 from Maxim Integrated is another ARM Cortex-M based processor suitable for the wearable market. It has a power consumption of $106\mu\text{A}/\text{MHz}$



Fig. 3: Focus is now shifting towards power-performance balance

in 600nA low-power mode. It sits in the higher edge of low-power processors. Clocked in at 96MHz , the 32-bit processor sports 512kB SRAM with 2MB flash. An interesting part of the processor is the trust protection unit with encryption and security features. These features include a modular arithmetic accelerator for fast ECDSA, a hardware PRNG entropy generator and a secure boot loader.

But it is not just about reducing power. As seen in MAX32630/1, removing encryption features could result in reduced power consumption, but the need for encryption is

prime in today's world. With every person holding multiple devices daily, securing data also becomes important. So a better fit, instead of just reducing power consumption, is to look at a scenario where the requirements are met with the lowest possible power.

Focus shifting towards power-performance balance

AMD Ryzen based on Zen architecture has been recently released. Apart from one processor model,

it focuses on power-performance balance. So, the move towards extracting the most out of the processor seems to be well underway. Reducing power consumption comes at a price, and it can only be done until a certain point, post which the performance dips below expectations.

We will always look towards processors

that consume low power to run more processes and allow extended battery life. Personally, getting an Intel or AMD processor to run off an AA battery would be the dream, but there is only so much that can be done.

For wearable devices, this is a whole different power struggle. With ARM Cortex M4 being the choice for major designs, future solutions should be interesting to look at. However, at present, balance between power consumption and performance looks to be the focus for processors of the future. **EPJ**

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